

highlights

September 16, 2026

1 Most important & most surprising results

1.1 Guitar domain for EasyStudy

A quick read of the headline findings, pulled from the two full notebooks in this repo (`guitar_evaluation.ipynb`, `guitar_interpretability.ipynb`). Every result below is copied straight from its source notebook - same code, same already-computed output - except the first chart, which restates an already-validated number as a picture instead of running it again here.

1.2 1. Combinations of neurons steer the model - single neurons don't

The core question behind the sparse-autoencoder work: is a *combination* of latent neurons more meaningful than any one neuron alone? The in-processing mechanism (`InProcessingSparseVAECF`) tests this directly by editing latent codes before decoding, not by re-ranking the output afterwards. Validated in `test_in_processing.py` (40 synthetic profiles) - see that script for the full run this chart summarizes.

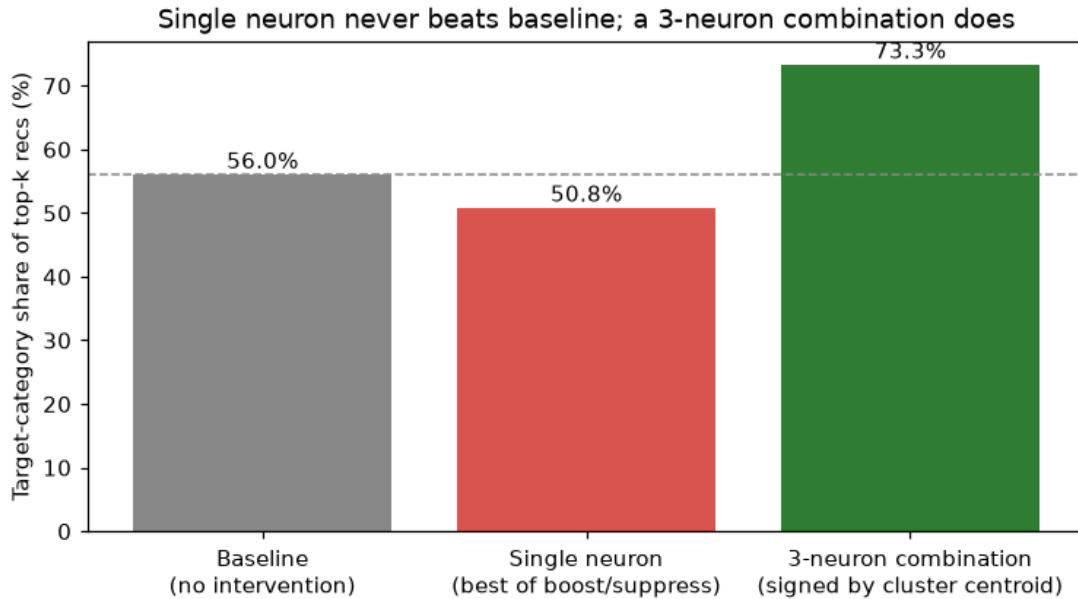
Result: nudging one neuron mostly just destabilizes the decode and never beats baseline. A correctly-signed *combination* of three neurons reliably shifts what gets recommended. This is live in the actual app too: a “Show me something different” button on the comparison page applies exactly this mechanism to a real participant’s session, not just a script.

```
[1]: # Summary of the in-processing intervention test (40 synthetic profiles,
# see test_in_processing.py for the full run this chart summarizes -
# reproduced here as a static chart rather than re-run, since the
# underlying test trains a fresh model and takes several minutes).
import matplotlib.pyplot as plt

conditions = ["Baseline\n(no intervention)", "Single neuron\n(best of boost/
↳ suppress)", "3-neuron combination\n(signed by cluster centroid)"]
target_share = [56.0, 50.8, 73.3] # % of top-k recs in the target category
colors = ["#888888", "#d9534f", "#2e7d32"]

fig, ax = plt.subplots(figsize=(7, 4))
bars = ax.bar(conditions, target_share, color=colors)
ax.set_ylabel("Target-category share of top-k recs (%)")
ax.set_title("Single neuron never beats baseline; a 3-neuron combination does")
ax.axhline(56.0, color="#888888", linestyle="--", linewidth=1)
for b, v in zip(bars, target_share):
```

```
ax.text(b.get_x() + b.get_width() / 2, v + 1, f"{v}%", ha="center")
plt.tight_layout()
plt.show()
```



(Source: *guitar_interpretability.ipynb*, section 2.5)

1.3 2.5. Neuron-*combination* interpretation

Section 2 only asks “which single neuron is an item most associated with?” One of the biggest questions behind this whole project is whether a *combination* of neurons carries more meaning than any single one alone.

Approach: cluster items by their *entire* latent-neuron association vector (KMeans over all LATENT_DIM dimensions of `item_neuron_matrix`, not any single column) with $k = 2 * \text{LATENT_DIM}$ clusters - each cluster centroid is then a genuine *combination* of neuron activations, not a single neuron in isolation. The same category/maker enrichment method from section 2 is then run per cluster, and compared against each cluster’s own best single contributing neuron.

```
[5]: from sklearn.cluster import KMeans

N_CLUSTERS = 2 * LATENT_DIM
kmeans = KMeans(n_clusters=N_CLUSTERS, n_init=10, random_state=42)
cluster_labels = kmeans.fit_predict(M)

cluster_summaries = []
for cluster_id in range(N_CLUSTERS):
    members = np.where(cluster_labels == cluster_id)[0]
```

```

if len(members) < MIN_COUNT:
    continue
    # Score each member by distance to its own cluster centroid
    ↪(tightest-fitting first)
    dists = np.linalg.norm(M[members] - kmeans.cluster_centers_[cluster_id],
    ↪axis=1)
    top = members[np.argsort(dists)][:TOP_N]

    cat_counts, maker_counts = Counter(), Counter()
    for i in top:
        cat_counts.update(categories_by_item[int(i)])
        maker_counts[maker_by_item[int(i)]] += 1
    cat, cat_ratio, cat_n = most_enriched(cat_counts, overall_cat_counts,
    ↪total_cats)
    maker, maker_ratio, maker_n = most_enriched(maker_counts,
    ↪overall_maker_counts, total_makers)

    # Which single neurons dominate this cluster's centroid, and what was
    # their own (section 2) enrichment? Lets us check if the combination
    # beats its strongest individual contributor.
    top_neurons = np.argsort(-np.abs(kmeans.cluster_centers_[cluster_id]))[:2].
    ↪tolist()
    best_contributor_ratio = max(
        (summary_df.loc[n, "category_enrichment"] for n in top_neurons if n in
    ↪summary_df.index),
        default=0.0,
    )

    cluster_summaries.append({
        "cluster": cluster_id, "size": len(members), "dominant_neurons":
    ↪top_neurons,
        "top_category": cat, "category_enrichment": round(cat_ratio, 1),
    ↪"category_n": cat_n,
        "top_maker": maker, "maker_enrichment": round(maker_ratio, 1),
        "beats_best_single_neuron": round(cat_ratio, 1) >
    ↪best_contributor_ratio,
    })

cluster_df = pd.DataFrame(cluster_summaries).set_index("cluster").
    ↪sort_values("category_enrichment", ascending=False)
display(cluster_df)

n_beating = cluster_df["beats_best_single_neuron"].sum()
print(f"\n{n_beating}/{len(cluster_df)} clusters show category enrichment
    ↪stronger than "
        f"either single dominant neuron's own enrichment from section 2.")

```

cluster	size	dominant_neurons	top_category \
11	211	[14, 7]	Beginner Kits
3	863	[14, 7]	Classical & Nylon-String Guitars
30	194	[15, 8]	Hollow & Semi-Hollow Body
7	210	[14, 0]	Hollow & Semi-Hollow Body
10	246	[14, 8]	Hollow & Semi-Hollow Body
2	268	[14, 8]	Hollow & Semi-Hollow Body
19	232	[8, 14]	Hollow & Semi-Hollow Body
12	371	[11, 14]	Electric Basses
28	356	[15, 14]	Beginner Kits
26	340	[14, 10]	Bass Guitars
0	251	[9, 14]	Bass Guitars
31	287	[14, 11]	Acoustic-Electric Guitars
9	265	[9, 14]	Acoustic-Electric Guitars
5	232	[3, 8]	Steel-String Acoustics
24	308	[0, 14]	Electric Basses
13	218	[0, 5]	Beginner Kits
21	309	[14, 7]	Solid Body
1	381	[14, 12]	Beginner Kits
20	347	[14, 12]	Beginner Kits
15	277	[12, 14]	Acoustic Guitars
18	360	[14, 11]	Acoustic-Electric Guitars
23	234	[3, 14]	Acoustic-Electric Guitars
8	313	[14, 7]	Acoustic-Electric Guitars
6	193	[12, 7]	Electric Basses
27	389	[5, 14]	Steel-String Acoustics
25	188	[11, 3]	Electric Basses
17	312	[7, 14]	Electric Basses
16	255	[11, 14]	Electric Basses
29	297	[14, 3]	Electric Basses
4	331	[8, 14]	Solid Body
22	217	[7, 12]	Solid Body
14	253	[5, 14]	Acoustic Guitars

cluster	category_enrichment	category_n	top_maker \
11	10.8	14	Best Choice Products
3	8.8	4	NaN
30	7.1	4	Grote
7	5.4	3	Fender
10	5.4	3	Gibson
2	5.4	3	AXL
19	5.4	3	NaN
12	4.2	5	NaN
28	3.9	5	NaN
26	3.8	5	Fender
0	3.8	5	Fender

31	3.7	4	Ibanez
9	3.7	4	YAMAHA
5	3.5	7	Fender
24	3.4	4	Ibanez
13	3.1	4	Fender
21	3.1	17	NaN
1	3.1	4	NaN
20	3.1	4	Dean Guitars
15	2.9	13	YAMAHA
18	2.8	3	YAMAHA
23	2.8	3	Dean Guitars
8	2.8	3	Fender
6	2.5	3	Fender
27	2.5	5	NaN
25	2.5	3	Directly Cheap
17	2.5	3	Fender
16	2.5	3	Fender
29	2.5	3	Takamine
4	2.0	11	Fender
22	2.0	11	Ibanez
14	1.6	7	NaN

	maker_enrichment	beats_best_single_neuron
cluster		
11	84.5	True
3	0.0	True
30	20.7	False
7	1.6	True
10	5.0	True
2	33.6	True
19	0.0	True
12	0.0	True
28	0.0	False
26	1.0	True
0	1.0	False
31	3.0	True
9	3.5	False
5	1.3	False
24	2.3	False
13	1.3	False
21	0.0	True
1	0.0	False
20	4.8	False
15	4.6	False
18	3.5	False
23	3.6	False
8	1.6	True
6	1.3	False

27	0.0	False
25	13.7	False
17	1.0	False
16	1.3	False
29	7.8	False
4	1.0	False
22	2.3	False
14	0.0	False

11/32 clusters show category enrichment stronger than either single dominant neuron's own enrichment from section 2.

If `n_beating` above is greater than zero, that's real evidence that some item groupings only show up when you look at a *combination* of neurons together, not any single one. If it's zero, the honest reading is that at this model size, the individual neurons already capture most of the interpretable structure, and combining them mostly just re-finds the same categories rather than revealing new ones - also a fair result, not a failure.

1.4 3. Best single-neuron result (for context)

(Source: *guitar_interpretability.ipynb*, section 2)

1.4.1 Inspect the single most interpretable neuron in detail

```
[4]: best_neuron = summary_df["category_enrichment"].astype(float).idxmax()
print(f"Most category-enriched neuron: {best_neuron} "
      f"({summary_df.loc[best_neuron, 'top_category']}, "
      f"{summary_df.loc[best_neuron, 'category_enrichment']}x base rate)")

top = np.argsort(-np.abs(M[:, best_neuron]))[:10]
for i in top:
    print(" -", loader.get_item_index_description(int(i)))
```

```
Most category-enriched neuron: 15 (Hollow & Semi-Hollow Body, 10.7x base rate)
- Fender FA-100 Limited Edition Dreadnought Acoustic Guitar with Gig Bag -
Black by Fender
- PRS Paul Reed Smith SE Standard 24 Electric Guitar, Tobacco Sunburst by PRS
Guitars
- Firefly FF338 Semi Hollowbody Guitar Ice Tea. by Firefly
- Firefly FFTH Semi-Hollow body Guitar Green color .(Black pearloid pickguard).
by Firefly
- Firefly FFJR Solid Body Electric Guitar (Metallic Gold). by Firefly
- Firefly FFTH Semi-Hollow body Guitar (Orange). by Firefly
- Firefly FFDCD Solid Body Electric Guitar Transparent Red) by Firefly
- Fender FA-100 Dreadnought Acoustic Guitar - Black Bundle with Gig Bag, Tuner,
Strings, Strap, and Picks by Fender
- Firefly FFTH Semi-Hollow body Guitar Sunburst Color(White pickguard). by
Firefly
```

- Leo Jaymz 6 Strings Headless Electric Guitar - Full Scale Travel Guitar - Ash Body and 5 piece Maple Neck - Customized Pickups (Brown) by Leo Jaymz

(Source: *guitar_interpretability.ipynb*, section 3)

1.5 3. “Give me something more surprising” mode

Same synthetic user profile, $\alpha=0$ (pure relevance, identical to plain VAE-CF) vs a high α (novelty-heavy re-rank) - using the *same* trained model’s weights, only the prediction-time blend changes.

```
[6]: # Build a synthetic profile from a handful of popular items so it's
      ↪reproducible without the elicitation module
item_popularity = loader.ratings_df.groupby("item").size().
      ↪sort_values(ascending=False)
seed_items = item_popularity.index[:30].to_numpy()
rng = np.random.default_rng(7)
profile = rng.choice(seed_items, size=3, replace=False).tolist()

print("Seed profile:")
for i in profile:
    print(" -", loader.get_item_index_description(int(i)))

def recommend_with_alpha(alpha, k=8):
    algo._alpha = alpha
    return algo.predict(selected_items=profile, filter_out_items=profile, k=k)

baseline = recommend_with_alpha(0.0)
surprising = recommend_with_alpha(0.9)

print("\n=== alpha=0.0 (pure relevance) ===")
for i in baseline:
    print(" -", loader.get_item_index_description(i))

print("\n=== alpha=0.9 (novelty-heavy / 'more surprising') ===")
for i in surprising:
    print(" -", loader.get_item_index_description(i))

overlap = len(set(baseline) & set(surprising))
print(f"\nOverlap between the two lists: {overlap}/{len(baseline)} items in_
      ↪common")
algo._alpha = 0.3 # restore the study's configured value
```

Seed profile:

- Piao 85115 Superior 112Db 330Hz and 400Hz Twin Tone Bass Horn, Black by Piao
- Ashthorpe Full-Size Cutaway Thinline Acoustic-Electric Guitar Package -
Premium Tonewoods - Blue by Ashthorpe
- Best Choice Products 41in Full Size Beginner All Wood Acoustic Guitar Starter
Set w/Case, Strap, Capo, Strings, Picks, Tuner - Blue by Best Choice Products

=== alpha=0.0 (pure relevance) ===

- Best Choice Products 38in Beginner All Wood Acoustic Guitar Starter Kit w/Case, Strap, Digital Tuner, Pick, Strings - SoCal Green by Best Choice Products
- Pyle Classical Acoustic Guitar Kit, 3/4 Junior Size Instrument for Beginner Kids, Adults, 36" Sun Burst by Pyle
- Donner Acoustic Guitar for Beginner Adult Full Size Cutaway Acustica Guitarra Bundle Kit with Free Online Lesson Bag Strap Tuner Capo Pickguard String Pick, Right Hand 41"Sunburst, DAG-1CS/DAD-160CS by Donner
- Davison Guitars Full Size Electric Guitar with 10-Watt Amp, Black - Right Handed Beginner Kit with Gig Bag and Accessories by Davison Guitars
- Best Choice Products 39in Full Size Beginner Electric Guitar Starter Kit w/Case, Strap, 10W Amp, Strings, Pick, Tremolo Bar - Jet Black by Best Choice Products
- Best Choice Products 22-Fret Full Size Acoustic Electric Bass Guitar w/4-Band Equalizer, Adjustable Truss Rod by Best Choice Products
- Pink Full Size Thinline Acoustic Electric Guitar With Free Gig Bag Case and Picks by Jameson Guitars
- Best Choice Products 41in Beginner Acoustic Guitar Full Size All Wood Cutaway Guitar Starter Set Bundle with Case, Strap, Capo, Strings, Picks, Tuner - SoCal Green by Best Choice Products

=== alpha=0.9 (novelty-heavy / 'more surprising') ===

- 30" Beginner Acoustic Guitar kids Guitar 1/2 Size Child Guitars Steel Strings with Gig Bag for Adult Girls Students by Strong Wind
- SX Left Handed 1/2 Size (34 inch) Electric Guitar Package Red with Amp, and Carry Bag SX RST 1/2 CAR LH by SX
- Fender DG-8S Solid Spruce Top Dreadnought Acoustic Guitar Pack with Gig Bag, Tuner, Strings, Picks, Strap, and Instructional DVD - Natural by Fender
- Oscar Schmidt OG2CE Dreadnought Acoustic Electric Guitar by Oscar Schmidt
- Fender FA-100 Limited Edition Dreadnought Acoustic Guitar with Gig Bag - Black by Fender
- Firefly FF338 Semi Hollowbody Guitar Ice Tea. by Firefly
- B.C. Rich Warlock Electric Guitar, Black by B.C. Rich
- Ibanez Performance PN1MHOPN Mahogany Parlor Acoustic Guitar High Gloss Natural by Ibanez

Overlap between the two lists: 0/8 items in common

(Source: *guitar_evaluation.ipynb*, section 5 - retrained on ~6x more data with bigger models)

1.6 5. Compute comparative metrics

- **Coverage:** fraction of the catalog that appears in at least one recommendation list across the synthetic users.
- **Novelty:** average popularity-complement (1 - fraction of users who interacted with the item) of recommended items - higher means the algorithm surfaces less mainstream

items.

- **Intra-list diversity (ILD):** average pairwise Jaccard *distance* between recommended items' category sets within each list - higher means a more varied single recommendation list.
- **Effect of Sparse VAE-CF's alpha:** it blends in a novelty term defined over *latent-neuron* profile dissimilarity (Part B's post-processing step) - that is a different notion of "different" than category-based ILD or popularity-based novelty computed below, so don't assume it must move both metrics in the same direction; treat this section as the empirical check of *whether/how* the post-processing changes the recommendations relative to plain VAE-CF, not as a confirmation of a predicted direction.

```
[6]: item_popularity = loader.ratings_df.groupby("item").size()
n_users = loader.ratings_df.user.nunique()
popularity_complement = (1.0 - item_popularity / n_users).reindex(all_items,
    ↪fill_value=1.0)

item_categories = {
    idx: set(loader.get_item_index_categories(int(idx))) for idx in all_items
}

def jaccard_distance(a, b):
    if not a and not b:
        return 0.0
    return 1.0 - len(a & b) / len(a | b)

def ild(rec_list):
    cats = [item_categories.get(i, set()) for i in rec_list]
    pairs = list(combinations(range(len(rec_list)), 2))
    if not pairs:
        return 0.0
    return np.mean([jaccard_distance(cats[i], cats[j]) for i, j in pairs])

rows = []
for name, recs in recommendations.items():
    flat = [item for rec_list in recs for item in rec_list]
    coverage = len(set(flat)) / len(all_items)
    novelty = np.mean([popularity_complement.get(i, 1.0) for i in flat])
    diversity = np.mean([ild(rec_list) for rec_list in recs])
    rows.append({"algorithm": name, "coverage": coverage, "novelty": novelty,
    ↪"ILD": diversity})

metrics_df = pd.DataFrame(rows).set_index("algorithm")
display(metrics_df)
```

	coverage	novelty	ILD
algorithm			
EASE (baseline)	0.114430	0.999542	0.558088
VAE-CF	0.015566	0.988391	0.565125
Sparse VAE-CF	0.009361	0.988208	0.477604

```
[7]: import matplotlib.pyplot as plt

fig, axes = plt.subplots(1, 3, figsize=(13, 4))
for ax, col in zip(axes, ["coverage", "novelty", "ILD"]):
    metrics_df[col].plot.bar(ax=ax, title=col)
    ax.tick_params(axis="x", rotation=30)
plt.tight_layout()
plt.show()
```

